

Microwave Oven CONTROL BOARD

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This project is designed to replace the defective control board in a microwave oven irrespective of brand and capacity. Microwave ovens are dumped as e-waste due to unserviceable control boards. Spare parts such as magnetron, HV transformer, HV diode and drive mechanism, except control board, are readily available in the market. Block diagram of the microwave oven is shown in Fig. 1

Circuit and working

The circuit diagram of the microwave oven is shown in Fig. 2. It is built around 5V voltage regulator 7805 (IC1), microcontroller (MCU) AT89C52 (IC2), four 12V relays (RL1 through RL4), five pnp transistors (BC557), 14 tactile switches, one 16x2 line LCD (LCD1) and a few discrete components.

The circuit is powered with 230V AC. Transformer X1 reduces voltage to 12 volts. Diodes D1 and D2 act as rectifier diodes. Capacitor C1 is connected as a filter. The rectified, filtered and regulated DC voltage is fed to the power section of the circuit, and 5V is fed to the control circuit via 7805.

AT89C52 MCU, running at a clock frequency of 11.0592MHz, is interfaced with 14 controlling switches, 16x2 LCD and four relays. Use Zener diodes ZD1 through ZD4 to connect 12V relay through a pnp transistor to 5V MCU. Diode D5 prevents the 12V supply to the MCU when the door switch is opened.

Relays RL1, RL2, RL3 and RL4

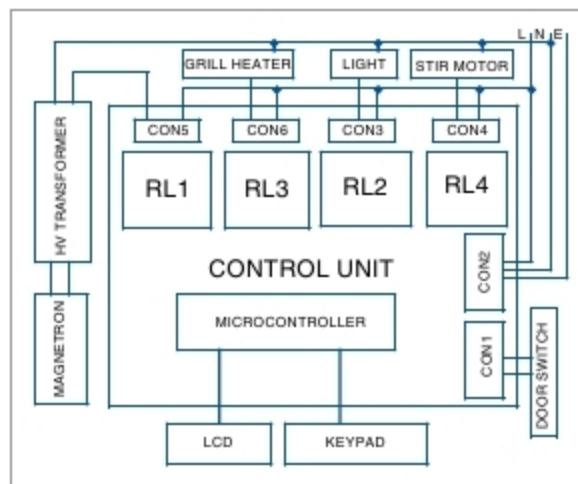


Fig. 1: Block diagram of microwave oven

are connected to the MCU through four relay driver transistors T1, T2, T3 and T4, respectively. Transistor T5 is used to drive the piezo buzzer. Pnp transistors are used as active low output from the controller. The LCD is connected at port P2. Potmeter VR1 can be used to adjust the contrast of LCD1. D4 through D7 are protection diodes for relay driver transistors. AT89C52 runs with a crystal frequency of 11.0592MHz.

Software

The software is written in C language and compiled with Kiel µvision 4 software. Logic behind the software is described briefly for easy understanding and editing. In the main part of the software, the MCU scans continuously the condition of the seven input switches, comprising vegetable (VEG), meat, rice, frozen (FRZ), bake, up and defrost (DFR). If any these switches are found de-pressed by the program, the software jumps

PARTS LIST	
Semiconductors:	
IC1	- 7805, 5V voltage regulator
IC2	- AT89C52 MCU
T1-T5	- BC557 pnp transistor
D1-D7	- 1N4007 rectifier diode
ZD1-ZD4	- 6.8V Zener diode
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2, R4, R5	- 5.6-kilo-ohm
R3	- 8.2-kilo-ohm
R6	- 100-ohm
R7-R10	- 10-kilo-ohm
VR1	- 10-kilo-ohm potmeter
Capacitors:	
C1	- 1000µF, 25V electrolytic
C2, C3	- 10nF ceramic disk
C4	- 100µF, 16V electrolytic
C5	- 10µF, 16V electrolytic
C6, C7	- 22pF ceramic disk
Miscellaneous:	
PZ1	- Piezo buzzer
S1-S14	- Tactile switch
LCD1	- 16x2 LCD
RL1, RL3	- 12V, 1CO, 30A relay
RL2, RL4	- 12V, 1CO, 1A relay
XTAL1	- 11.0592MHz crystal
CON1, CON3-CON6	- 2-pin connector
CON2	- 3-pin connector
X1	- 230V AC primary to 12V-0-12V, 500mA secondary transformer

to the corresponding sub-routine. On auto sub-routine, each branch sets the total time of cooking and on/off delay timing for magnetron and grill heater. Total delay and on/off delay against each cooking styles are shown in the table.

On de-pressing Defrost, the program executes defrost sub-routine. Defrosting technique provided in the microwave oven is for cooking or heating refrigerated food. Normal cooking of refrigerated food in an oven results in rapid burning of outer parts, resulting in damaged food.

For compensating this at defrost mode, the system initially switches on the magnetron just for a few seconds. It slowly increases on time and decreases off time. This technique helps penetrate the heat throughout the food evenly.

Grill option is provided for meat only. Combi option is provided for

TOTAL DELAY FOR DIFFERENT COOKING STYLES

Mode	Veg		Meat		Rice		Bake	Defrost
Style	Micro	Combi	Micro	Grill	Combi	Micro	Combi	Micro
Micro On (Sec)	12	12	15	Off	15	12	12	10
Micro Off (Sec)	8	8	5	Off	5	8	8	10
Grill On (Sec)	Off	8	Off	22	5	Off	8	10
Grill Off (Sec)	Off	12	Off	8	15	Off	12	10
Total (Min)	5	3	15	15	10	10	6	20
Cycles	15	9	45	45	30	30	18	60